MILAN TIWARI

M.S. Robotics and Autonomous Systems — May 2026 Graduate
(602) 471-4821 — milantiwari890@gmail.com — linkedin.com/in/milan-tiwari-890 — github.com/milan-tiwari

PROFILE

Graduate student in Robotics and Autonomous Systems with experience in Embodied AI, Computer Vision, 3D Scene Understanding, Deep Learning, and Autonomous Systems. Skilled in developing intelligent robotic systems, perception pipelines, and AI-driven applications from research prototypes to deployable solutions.

EDUCATION

- **Arizona State University**, Tempe, AZ — Aug 2024 – May 2026
  M.S. in Robotics and Autonomous Systems — CGPA: 3.45/4.00
  Focus: Embodied AI, Computer Vision, 3D Scene Understanding, Autonomous Systems
- **Jagran Lakecity University**, Bhopal, India — Aug 2019 – Jul 2023
  B.Tech. (Hons) CSE (AI) — CGPA: 3.57/4.00

PROJECTS

ProteinNet-Hybrid — Sequence-to-Structure Inference

- Built a Transformer encoder in PyTorch DDP consuming amino-acid sequence and evolutionary-profile (PSSM) inputs from ProteinNet CASP12, achieving C-alpha coordinate prediction at 10.15 Å validation RMSD after 100 epochs of distributed training.
- Engineered a two-mode inference system (FASTA-only and MSA alignment), packaged into a published Docker image with bundled checkpoint and config for zero-setup CPU inference and reproducible PDB/JSON structure export.
- Evaluated exported predictions through PyMOL overlay inspections across 10 CASP12 targets, yielding an aligned RMSD range of 2.43–15.59 Å and mean of 9.17 Å, confirming reliable coarse fold capture.

Multi-Robot Disaster Management — Safety-Aware Relief Coordination

- Designed a MATLAB multi-robot disaster-response simulator combining weighted-Voronoi coverage control over a spatial priority and time-varying demand field with greedy unique hotspot claiming to eliminate duplicate task assignment across robots.
- Implemented a CBF-style safety controller with QP-based primary filtering and smooth pairwise repulsion fallback, paired with a capacity-aware service model that depletes demand within a service radius and triggers base-return refill cycles.
- Instrumented full metric tracking across locational cost, unmet demand, minimum inter-robot separation, and served coverage over a 180 s scenario, enabling quantitative evaluation of fleet coordination and safety margins.

LERF-TOGO / Gaussian Splatting — Zero-Shot 3D Grasp Semantics

- Extended the LERF-TOGO grasping pipeline with a Gaussian Splatting scene backend abstraction supporting both original LERF/Nerfstudio and splatfacto-based geometry from a unified grasp-generation interface.
- Upgraded scene semantics from CLIP to SigLIP 2 multi-scale pyramid embeddings over HLOC/COLMAP-reconstructed tabletop scenes, with robot-frame alignment to transfer SfM geometry into executable robot coordinates.
- Integrated GraspNet/Dex-Net collision-filtered grasp generation over world-frame point clouds, served through an interactive Viser UI supporting natural-language object-part queries for zero-shot semantic manipulation.

PROFESSIONAL EXPERIENCE

Data Storytelling Assistant — *Newswell (Arizona State University), Tempe, Arizona* — Nov 2025 – Present

- Built and owned a Python ingestion and summarization pipeline across RSS/local-news feeds, Reddit/community signals, and Hyland government agenda portals, generating daily editorial digests from public source data.
- Expanded government-meeting ingestion beyond raw PDFs by parsing meeting pages, agenda links, supporting documents, item-level detail pages, and available YouTube transcripts for richer civic-news summaries.
- Implemented OpenAI-based summarization and digest-generation workflows that synthesize ingested items into reporter-ready local-news briefs, government updates, concise editorial overviews, and HTML/TXT outbox artifacts.
- Migrated the workflow from local SQLite/cron execution to a Dockerized Azure production stack using Azure Container Apps Jobs, Azure Container Registry, Azure Database for PostgreSQL, Key Vault secrets, Azure Files outbox storage, and Log Analytics.

Software Engineer — *Patel Motors (Indore) Pvt. Ltd., Indore, Madhya Pradesh* — May 2023 – May 2024

- Built an ML-assisted Flask + SQL quotation engine that automated client quote generation by learning from historical pricing patterns, replacing manual assembly with faster, data-driven, and fully traceable response handling.
- Rebuilt the company website with HTML, CSS, JavaScript, and Flask-backed forms, integrating dynamic recommendation logic to surface relevant vehicle options based on customer inquiry inputs.
- Designed and deployed a lightweight customer intent classifier on incoming inquiry data, enabling automated lead prioritization and routing to the appropriate sales workflow.

Machine Learning Intern — *Value Matrix, Dover, Delaware* — Sep 2022 – Nov 2022

- Engineered core components of a real-time eye-movement tracking pipeline using computer vision models, redesigning the evaluation workflow to improve analysis accuracy by 20
- Built an end-to-end review interface integrating AWS Transcribe for automated speech-to-text transcription with a Flask + JavaScript frontend, cutting turnaround time from model output to analyst review significantly.
- Researched and benchmarked multiple gaze estimation architectures to identify the highest-performing model configuration, informing pipeline design decisions and improving overall system reliability.

Associate Intern — *Feynn Labs, Gauhati, Assam* — Jul 2022 – Sep 2022

- Built EV market-segmentation analyses from customer and competitor data, enabling reports that identified high-potential customer groups for business strategy.

SKILLS

Languages: Python, C++, SQL, JavaScript, MATLAB, C.
Robotics & Frameworks: ROS, PyTorch, OpenCV, Nerfstudio, Gazebo, TensorFlow, TensorFlow Lite, Flask.
AI/ML Concepts: Embodied AI, Gaussian Splatting, Zero-Shot Grasping, Computer Vision, LLMs, Reinforcement Learning, Deep Learning, Autonomous Systems.
Tools & Infrastructure: Git, Docker, Azure, Azure Container Apps, Azure Container Registry, Azure Key Vault, PostgreSQL, SQLite, AWS, Hugging Face, MATLAB Simulink.